

Private Encrypted Data Alignment by ID

A design note: extending OKVS/RLWE Reverse-PMT to payload transport

Working notes

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Abstract

We sketch how to repurpose the OKVS-based, RLWE-instantiated Reverse-PMT (RPMT) of the companion paper into a *private encrypted data alignment* primitive: a receiver R holds a table $\{(ID_i, \text{payload}_i)\}$ and a sender S holds a set of IDs; for each of S 's IDs that matches one of R 's, S obtains an *encryption* of the corresponding payload, aligned to S 's own order, so that S can run a homomorphic analysis on the aligned payloads together with its own data. The two non-trivial pieces are (i) neutralizing the OKVS “garbage” on non-matching IDs without leaking the join pattern, via a secret-shared membership indicator and a single ciphertext $\times$ ciphertext mask, and (ii) carrying real-valued payloads in CKKS, which turns the OKVS encoding problem from a finite-field full-rank question into a *numerical conditioning* question. We record the design decisions and the resulting research questions.

1 Setting and target functionality

Let R (receiver, data owner) hold a key–value table $T = \{(ID_i, \text{payload}_i)\}_{i \in [n_x]}$ and let S (sender) hold a set of identifiers $\{ID_j^S\}_{j \in [n_y]}$. We want a protocol that delivers to S , for each j , an *encryption* of payload_i whenever $ID_j^S = ID_i$ for some i , and an encryption of (effectively) zero otherwise; the result is aligned to S 's own indexing. S then evaluates an application-specific homomorphic function on these aligned ciphertexts and its own plaintext data. Throughout, R holds the RLWE secret key(s); all ciphertexts handed to S are under R 's public key, so S computes but cannot decrypt. The final analysis output is returned to R (or kept secret-shared); this delivery and the choice of analysis are treated as out of scope here.

2 Recap of the base RPMT mechanism

The base construction (companion paper) realizes F_{RPMT} : $b_j = \mathbf{1}(y_j \in X)$. Its mechanics that we reuse:

1. R OKVS-encodes $\{(x'_i, \text{IS})\}$ — *same* indicator value IS for every key — into p , using the *Random Spaced Band* (RSB) matrix (a fixed column permutation π_N of a random-band matrix, aligning bands to the RLWE slot size N).
2. R RLWE-encrypts p (block-packed, plus pre-rotated copies) and sends it.
3. S homomorphically evaluates the OKVS decoding $M \cdot p$ as a (block-)diagonal matrix–vector product, using only slot-wise PlainMult/CtxtAdd. *Sequencing* (Algorithm 3 of the paper) splits the collision-laden matrix into L layers $\bar{M}_t \in \mathbb{Z}_p^{N \times m}$, each with $\leq w + \zeta$ diagonals; the assignment

of item i to layer/slot (μ_i, τ_i) depends only on the band start $a_i = H_1(k_i)$, the slot count N , and the span limit ζ — *not* on w .

4. Each decoded slot is **IS** if $y_j \in X$ and a pseudorandom value otherwise. The base protocol re-randomizes, returns the L ciphertexts to **R**, who tests equality with **IS**.

A key efficiency point: sequencing is an n -balls-into- N -bins process, so its batching efficiency relies on the regime $N \ll n$ (well-filled slots). The base parameters fix $N = 2^{13}$ for all $n \in [2^{16}, 2^{22}]$.

Secret-shared PMT (Appendix D). Masking the decoded plaintext by r_μ before **R** decrypts, followed by a secret-shared equality test $F_{\text{ss-peqt}}$, yields XOR shares $b_{X,i}$ (held by **R**) and $b_{Y,i}$ (held by **S**) with $b_{X,i} \oplus b_{Y,i} = \mathbf{1}(y_i \in X)$. We reuse this verbatim for the indicator.

3 The alignment construction

The construction runs *two* OKVS instances on the same OPRF'd keys $k_i = \text{OPRF}(\text{ID}_i)$: an *indicator* OKVS over a finite field (all values = **IS**, exactly the base protocol, used only to obtain membership shares), and a *payload* OKVS whose value for k_i is payload_i , decoded homomorphically so that **S** holds a ciphertext whose slot for y_j contains payload_i on a match and pseudorandom garbage on a non-match. A generic payload no longer self-certifies membership (garbage is indistinguishable from a real payload), which is exactly why the separate indicator is needed: it provides a secret-shared membership bit with which the garbage is masked off. The receiver-side encoding is Algorithm 1; the sender-side alignment-and-masking is Algorithm 2.

Algorithm 1 II.Encode — receiver **R**, on table $\{(\text{ID}_i, \text{payload}_i)\}_{i \in [n_x]}$

- 1: $k_i \leftarrow \text{OPRF}(\text{ID}_i)$ for all i ▷ OPRF preprocessing (**R** is OPRF receiver)
 - 2: **IS** $\leftarrow$ random indicator tag
 - 3: $p^{\text{ind}} \leftarrow \text{OKVS.ENCODE}_{\mathbb{Z}_p}(\{(k_i, \text{IS})\}_i)$ ▷ indicator OKVS, finite field
 - 4: $p^{\text{pay}} \leftarrow \text{OKVS.ENCODE}_{\mathbb{R}}(\{(k_i, \text{payload}_i)\}_i)$ ▷ payload OKVS, min-norm solve (Alg. 3)
 - 5: $C^{\text{ind}} \leftarrow \text{BGV.Enc}(p^{\text{ind}})$, $C^{\text{pay}} \leftarrow \text{CKKS.Enc}(p^{\text{pay}})$ ▷ block-packed
 - 6: **send** $C^{\text{ind}}, C^{\text{pay}}$ to **S** ▷ (rotation/Galois keys for the spaced-band decode are sent in setup)
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Algorithm 2 II.Align — sender **S**, on set $\{y_j\}_{j \in [n_y]}$ (and **R**'s share step)

- 1: $(c_\mu^{\text{ind}})_\mu \leftarrow \text{HOMDECODE}(C^{\text{ind}}, \{y_j\})$ ▷ base-protocol indicator decode
 - 2: $(c_\mu^{\text{pay}})_\mu \leftarrow \text{HOMDECODE}(C^{\text{pay}}, \{y_j\})$ ▷ CKKS spaced-band decode, Alg. 4
 - 3: run ssPMT on $(c_\mu^{\text{ind}})_\mu$: **S** gets share $b_{Y,j}$, **R** gets $b_{X,j}$, with $b_{X,j} \oplus b_{Y,j} = \mathbf{1}(y_j \in X)$
_____ garbage removal (deferred; see Section 4 note) _____
 - 4: **R sends** $\text{Enc}_{\text{CKKS}}(b_{X,j})$, packed into the payload slot layout
 - 5: **S** sets $s_j \leftarrow b_{Y,j}$ and forms $\text{Enc}(b_j) \leftarrow \text{PlainAdd}(\text{PlainMult}(\text{Enc}(b_{X,j}), 1 - 2s_j), s_j)$ ▷ XOR by a *plaintext* bit: no $\text{ctxt} \times \text{ctxt}$
 - 6: **S** computes $\text{Enc}(\widetilde{\text{payload}_j}) \leftarrow \text{Enc}(b_j) \odot c_{(\cdot)}^{\text{pay}}[j]$ ▷ the single $\text{ctxt} \times \text{ctxt}$ mult; $1 \cdot \text{payload}$ or $0 \cdot \text{garbage}$
 - 7: **return** aligned $\{\text{Enc}(\widetilde{\text{payload}_j})\}_j$ ▷ **S** runs its analysis; circuit-privacy/noise-flood before **R** decrypts
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Two facts about Algorithm 2. The XOR in line 5 uses $a \oplus b = a + b - 2ab$ (not $a + b - ab$, which is OR); since **S**'s share s_j is a *plaintext* bit, line 5 is a plaintext multiply-add, with

no ciphertext×ciphertext product. Line 6 is the *only* ciphertext×ciphertext multiplication — it takes the protocol out of the base paper’s additive-only regime by one multiplicative level (plus relinearization), to be budgeted in the modulus. The membership bit b_j never appears in the clear to either party, so the join pattern stays hidden from S.

Remark 1 (circuit privacy). Since S’s computation touches the ciphertext before R decrypts the final analysis output, noise-flooding / circuit privacy must be applied before that decryption, exactly as the base protocol does on its returned ciphertexts.

3.1 Scheme split: indicator in BGV/BFV, payload in CKKS

Real-world payloads are typically real numbers, so the payload OKVS is naturally carried in CKKS while the indicator/equality logic stays in BGV/BFV. The re-encryption of $b_{X,i}$ into CKKS is free: the share is plaintext to its holder, so no scheme-switching primitive is required — R simply CKKS-encrypts a fresh 0/1, and the plaintext-bit XOR of Algorithm 2 (line 5) runs in CKKS. The only semantic change from the BGV masking is that *exact* zeroing becomes *approximate*: with $b_i = 0 \pm \delta$, a non-match leaves a residual $\delta \cdot \text{garbage}$. Hence the central numerical requirement:

$$\|p\| \leq \frac{\|v\|}{\sigma_{\min}(M)}, \quad |\text{garbage}| = |\langle \text{row}(y'), p \rangle| \leq \sqrt{w} \|p\|, \quad \text{CKKS residual} \sim \delta \cdot |\text{garbage}|,$$

so controlling $\sigma_{\min}(M)$ (equivalently $\|p\|$) is what keeps the residual below the analysis tolerance (Section 4).

3.2 Slot geometry and the “promote later” parameter plan

BGV/BFV with ring dimension N has N slots; CKKS with ring dimension N' has $N'/2$ slots. To share one sequencing grid between the two decodes (so R packs its membership share once, with no cross-grid permutation), the slot counts must match. The clean way is to choose the CKKS ring as $N' = 2N$, which CKKS likely needs anyway for the masking depth and payload precision. Then both have N slots and the (μ_i, τ_i) layout is identical (it depends only on H_1 , the slot count, and ζ ; w and the band values may differ between the two instances).

Two further points drive the parameterization:

- **Keep N small during alignment.** Sequencing is n -balls-into-(slot count); large slot counts under-fill the bins and waste computation on near-empty ciphertexts. The alignment phase should run at modest N (e.g. BFV 2^{13} / CKKS 2^{14}), preserving $N \ll n$.
- **Bootstrap is not a dimension lift.** N is fixed at key generation; bootstrapping refreshes the modulus/level at the *same* N . If deep analysis is needed, one keeps alignment at small N and, *after* masking (alignment done), ring-switches the aligned payload *up* to a large ring and bootstraps there (a sparse-slot, hence favorable, bootstrapping). The big- N bootstrapping keys must be provisioned upfront, but the shallow alignment work stays cheap at small N . Ring switching *down* (large→small) is the genuinely useful direction for shrinking the tail; ring switching *up* grants no fresh noise budget by itself — only the subsequent bootstrap does.

4 Real-field OKVS: conditioning instead of rank

For the payload OKVS over $\mathbb{R}$ (CKKS) we use *Gaussian* random bands: each width- w band has i.i.d. $\mathcal{N}(0, 1)$ coefficients, placed at a random start. The encoding solves $Mp = v$ over $\mathbb{R}$, where M is $n \times m$, $m \approx (1 + \varepsilon)n$.

Why Gaussian, not binary. The earlier motivation for 0/1 bands was an “addition-only” homomorphic decode, but this is illusory: the decode is $\sum_k \text{PlainMult}(\rho_k(c), \text{diag}_k)$, and a **PlainMult** costs the same whether the plaintext diagonal is 0/1 or real (it is encoded into a dense plaintext polynomial and NTT-multiplied regardless); binary also gives no encoding-time speedup off GF(2). Meanwhile full-entropy real coefficients condition markedly better — experimentally $\sigma_{\min}$ is $\sim 1.7\times$ larger at equal w , so Gaussian bands reach a target $\sigma_{\min}$ with $\sim 1/3$ fewer diagonals — and admit clean random-matrix tools (continuous anti-concentration) for the analysis below.

Rank is automatic; conditioning is the whole question. For Gaussian bands full row rank holds almost surely (the only structural requirement is that the sparsity pattern admits a row-to-column matching, which holds once (ε, w) clear the RB-OKVS thresholds). So unlike the finite-field case there is no rank-failure event to bound; CKKS instead needs $\sigma_{\min}(M)$ bounded *away from* 0, since a full-rank M with tiny $\sigma_{\min}$ already yields an exploding $\|p\|$. The operative target is a *smallest-singular-value tail bound*

$$\Pr[\sigma_{\min}(M) < t] \leq 2^{-\lambda},$$

which replaces the finite-field rank-failure bound and which sets (ε, w) ; a proof sketch is given in Section 5. Redundancy ε (extra columns) is the primary knob — it sets the spectral gap $\sigma_{\min} \gtrsim \sqrt{m} - \sqrt{n}$ — with w secondary, as the experiments confirm.

Efficient minimum-norm encoding. Among all valid encodings $\{p_0 + \ker M\}$ the minimum- ℓ_2 -norm solution is $p^* = M^\top(MM^\top)^{-1}v$, and its size is governed precisely by $\sigma_{\min}$: $\|p^*\| \leq \|v\|/\sigma_{\min}(M)$. Crucially this is cheap to compute for band matrices: ordering rows by band start makes $MM^\top$ *banded* (bandwidth $O(w)$), so $MM^\top y = v$ is solved by banded Cholesky in $O(nw^2)$ and $p^* = M^\top y$ in $O(nw)$ — comparable to the encoding itself, and far smaller in norm than the naive triangular particular solution (free variables set to 0), whose norm is governed by the worse-conditioned pivot submatrix. (Normal equations square the condition number; a banded QR of $M^\top$ is the stable fallback.) The encoding and the spaced-band homomorphic decode are Algorithms 3 and 4.

Algorithm 3 OKVS.ENCODER_ℝ — minimum-norm real-field OKVS encoding

Require: keys $\{k_i\}_{i \in [n]}$, values $v \in \mathbb{R}^n$, width w , expansion ε ($m = \lceil(1 + \varepsilon)n\rceil$)

- 1: for each i : start $a_i \leftarrow H_1(k_i) \in [0, m-w]$; in-band coeffs $(M_{i,a_i+j})_{j < w} \leftarrow H_2(k_i) \sim \mathcal{N}(0, 1)$
 - 2: sort rows by a_i and form the banded SPD Gram matrix $G = MM^\top$ (bandwidth $O(w)$)
 - 3: $G = LL^\top$ ▷ banded Cholesky, $O(nw^2)$; SPD $\Rightarrow$ pivot-free, stable
 - 4: solve $LL^\top y = v$ by two banded triangular solves ▷ $O(nw)$
 - 5: **return** $p^* = M^\top y$ ▷ min-norm solution; $\|p^*\| \leq \|v\|/\sigma_{\min}(M)$
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Remark 2 (implementation status). The accompanying CKKS implementation realizes Algorithms 3 and 4: the encode/decode round-trip that delivers $\langle \text{row}(y_j), p \rangle$ in the encrypted state, including the multi-block N -spacing ($m = bN$) with Galois-key homomorphic rotations for the super-diagonals (rotation by r computed once per block, reused across layers), verified correct for all queries at scale 2^{40} . Deferred: the *clustered* sequencing of Algorithm 3 of the base paper (the current sequencing is naive, one query per slot per layer), and masking off the non-match garbage (lines 4–6 of Algorithm 2, needing the indicator OKVS and the ciphertext $\times$ ciphertext multiply).

Algorithm 4 HOMDECODECKKS — spaced-band decode, rotation-key variant

Require: CKKS ciphertexts $(c_0, \dots, c_{b-1})$ of $p = (p_0 \| \dots \| p_{b-1})$, $p_i \in \mathbb{R}^N$ (N slots, $b = m/N$); query keys $\{y_j\}$; Galois (rotation) keys

- 1: build the RSB query matrix M' from $\{y_j\}$ (N -spaced bands); diagonalize and sequence into layers $\{\bar{M}_t\}$, each with offset set $D_t \subseteq \{0, \dots, w+\zeta-1\}$
- 2: **for** each distinct offset $k \in \bigcup_t D_t$ **do**
- 3: $\rho_k \leftarrow \text{HOMROTATE}(c_\bullet, k)$ $\triangleright$ homomorphic rotation via Galois key, computed *once*, reused across layers
- 4: **for** each layer t **do**
- 5: $c_t^* \leftarrow \sum_{k \in D_t} \text{PlainMult}(\rho_k, \text{diag}_{t,k})$ $\triangleright w+\zeta$ slot-wise plain-mult/adds
- 6: **return** $\{c_t^*\}_t$ $\triangleright$ slot of y_j holds $\langle \text{row}(y_j), p \rangle = \text{payload (match) or garbage (non-match)}$

5 The $\sigma_{\min}$ tail bound

Model. $M \in \mathbb{R}^{n \times m}$, $m = \lceil (1 + \varepsilon)n \rceil$; row i is supported on $[a_i, a_i + w)$ with $a_i \sim U[0, m - w]$ i.i.d., and the w in-band entries are i.i.d. $\mathcal{N}(0, 1)$. We write $r_1, \dots, r_n$ for the rows.

Theorem 1 (Conditioning of Gaussian band matrices). *Fix $\varepsilon_0 \in (0, 1]$. There are constants $c_1, c_2, c_3 > 0$ depending only on ε_0 such that for every n , every $\varepsilon \geq \varepsilon_0$, and every width $c_1 \log n \leq w \leq n$, the model above satisfies*

$$\Pr [\sigma_{\min}(M) < c_2] \leq e^{-c_3 \min(\varepsilon n, w)}.$$

More precisely there is a threshold $t(\varepsilon, w)$, non-decreasing in both arguments and bounded below by $c_2 > 0$, with $\Pr[\sigma_{\min}(M) < t(\varepsilon, w)] \leq e^{-c_3 \min(\varepsilon n, w)}$. In particular $\sigma_{\min}(M)$ is bounded away from 0 by a constant independent of n with probability $1 - e^{-\Omega(\min(\varepsilon n, w))}$: the conditioning does not degrade with set size.

The bound is governed by $\min(\varepsilon n, w)$. For the set sizes of interest $\varepsilon n \gg w$, so the binding term is w : $\log \Pr[\text{ill-conditioned}] \approx -c_3 w$ is linear in w , and a target failure $2^{-\lambda}$ needs $w = \Omega(\lambda)$ — the prediction tested empirically in the companion experiment. The next corollary turns the $\sigma_{\min}$ bound into the two quantities the construction actually needs: a bounded encoding and a bounded encode/decode precision loss.

Corollary 1 (Bounded encoding and round-trip precision). *Under Theorem 1, except with probability $e^{-c_3 \min(\varepsilon n, w)}$:*

(i) *the min-norm encoding obeys $\|p^*\| \leq \|v\|/c_2$, a bound independent of n (so the CKKS garbage $|\langle \text{row}(y'), p^* \rangle| \leq \sqrt{w} \|p^*\|$ is likewise $O(\sqrt{w} \|v\|)$);*

(ii) *if S decodes from the CKKS ciphertext of p^* held with error $\tilde{p} = p^* + e$, $\|e\|_2 \leq \eta$, then every matching query $y_j = x_i$ recovers*

$$v'_j = \langle \text{row}(y_j), \tilde{p} \rangle = v_i + \langle \text{row}(y_j), e \rangle, \quad |v'_j - v_i| \leq \|\text{row}(y_j)\|_2 \eta = \Theta(\sqrt{w}) \eta.$$

Hence preserving b bits of payload precision through the encode/decode round-trip costs $\approx \frac{1}{2} \log_2 w + \log_2(1/\sigma_{\min}) = O(\log(\sqrt{w}/\sigma_{\min}))$ extra bits of CKKS scale — the $1/\sigma_{\min}$ factor is the encoding's norm inflation, the $\sqrt{w}$ the band-sum amplification.

Proof of Corollary 1. (i) is $\|p^*\| \leq \|v\|/\sigma_{\min}(M) \leq \|v\|/c_2$ on the high-probability event of Theorem 1, and $|\langle \text{row}(y'), p^* \rangle| \leq \|\text{row}(y')\| \|p^*\|$ with $\|\text{row}(y')\| = \Theta(\sqrt{w})$. (ii): for a matching key $\text{row}(y_j) = \text{row}(x_i)$, so $\langle \text{row}(y_j), p^* \rangle = v_i$ by OKVS correctness; the error is $\langle \text{row}(y_j), e \rangle$, bounded by Cauchy–Schwarz. The bit count follows by setting $\sqrt{w}\eta$ below the payload LSB and noting η scales with $\|p^*\|$, which carries the $1/\sigma_{\min}$ factor. $\square$

Proof sketch (full proof in Appendix A). Write $\sigma_{\min}(M) = \min_{u \in S^{n-1}} \|M^\top u\|_2$, $M^\top u = \sum_i u_i r_i$. The band structure makes the m coordinates of $M^\top u$ *independent* Gaussians of total variance w (distinct columns use disjoint entries: $(M^\top u)_j \sim \mathcal{N}(0, s_j^2(u))$, $s_j^2(u) = \sum_{i: j \in [a_i, a_i+w)} u_i^2$, $\sum_j s_j^2 = w$). Tensorizing the per-coordinate Gaussian anti-concentration gives a single-direction small ball $(C\rho)^{\kappa(u)}$, with $\kappa(u) = \#\{j : s_j^2(u) > 0\}$ the number of *active columns* the direction touches. One then splits S^{n-1} : *incompressible* directions spread over $\kappa = \Theta(\varepsilon n)$ columns (beating the $e^{O(n)}$ spherical net), *compressible* ($\approx k$ -sparse) ones still touch $\kappa = \Omega(w)$ columns (beating the small k -sparse net once $w = \Omega(\log n)$); random starts make the pattern well-spread except with probability $e^{-\Omega(w)}$ (balls-into-bins). Together these give the $\min(\varepsilon n, w)$ exponent. Thus ε sets the conditioning scale (spectral gap) and the incompressible exponent, w the compressible exponent — matching the experiment, where $\sigma_{\min}$ rises with both and collapses as $\varepsilon \rightarrow 0$. Gaussian coefficients make the tensorized step the clean Gaussian small ball; binary would need a Carbery–Wright / Berry–Esseen replacement.

6 Open problems / next steps

Open problem 1 (sharpen the $\sigma_{\min}$ threshold). Theorem 1 (proof in Appendix A) gives the qualitative tail $\Pr[\sigma_{\min} < c_2] \leq e^{-c_3 \min(\varepsilon n, w)}$. What remains is the *tight* threshold $t(\varepsilon, w)$ — an explicit $(\varepsilon, w) \mapsto t$ matching the measured surface (and hence the smallest w for a target $\sigma_{\min}$). The compressible-net balancing (Appendix Step B.3) and the optimal constant in the spectral gap are the parts needing care.

Open problem 2 (empirical conditioning). Measure, for Gaussian band matrices over $\mathbb{R}$: the amplification $\|p\|/\|v\|$ for naive-triangular vs. banded-min-norm; $\sigma_{\min}(M)$ and its dependence on (w, ε) ; and the resulting garbage magnitude $|\langle \text{row}(y'), p \rangle|$ on non-matches. (Addressed by the accompanying experiment, which also confirms Gaussian bands beat binary at equal w .)

Open problem 3 (2-row decode). The $2 \times (N/2)$ BGV hypercube admits, with $N/2$ -based sequencing and rotation-during- compute, two layer products per row-wise rotation — halving plain-mults for the indicator decode and resolving the open remark of §4.3 of the base paper. The packing (offset alignment of paired layers; replication vs. windowed layout) and its communication/computation trade-off remain to be fixed.

A Full proof of Theorem 1

Throughout, c, C, c_i, C_i denote positive constants depending only on ε_0 , and we condition on the starts $\{a_i\}$ until Step B.4. Recall $\sigma_{\min}(M) = \min_{u \in S^{n-1}} \|M^\top u\|_2$ and $M^\top u = \sum_i u_i r_i$.

B.1 Independent-coordinate representation. For $u \in S^{n-1}$ the coordinates of $M^\top u$ are $(M^\top u)_j = \sum_{i: j \in [a_i, a_i+w)} u_i M_{ij}$. Entries $\{M_{ij}\}$ with distinct j are disjoint sets of i.i.d. $\mathcal{N}(0, 1)$ variables, so $\{(M^\top u)_j\}_{j \in [m]}$ are independent with $(M^\top u)_j \sim \mathcal{N}(0, s_j^2(u))$, $s_j^2(u) = \sum_{i: j \in [a_i, a_i+w)} u_i^2$,

and $\sum_j s_j^2(u) = w\|u\|_2^2 = w$. Let $\kappa(u) = \#\{j : s_j^2(u) > 0\} = |\bigcup_{i \in \text{supp}(u)} [a_i, a_i + w]|$ be the number of active columns.

Lemma 1 (tensorized small ball; Rudelson–Vershynin). *There are constants $C_0, p_0 < 1$ such that for every fixed u and every $\rho \leq 1$,*

$$\Pr[\|M^\top u\|_2 \leq \rho\sqrt{w}] \leq (C_0\rho)^{\kappa(u)/2}.$$

Proof. Each Gaussian coordinate satisfies $\Pr[|(M^\top u)_j| \leq \rho s_j(u)] \leq \sqrt{2/\pi}\rho$. Restricting to the $\kappa(u)$ active coordinates and applying the tensorization lemma of Rudelson–Vershynin (independent coordinates, uniform single-coordinate small ball) gives $\Pr[\sum_j (M^\top u)_j^2 \leq \rho^2 \sum_j s_j^2(u)] \leq (C_0\rho)^{\kappa(u)/2}$; substitute $\sum_j s_j^2(u) = w$. $\square$

B.2 Good sparsity pattern. Call the pattern $\{a_i\}$ *good* if (a) every column lies in $O(w/(1+\varepsilon))$ bands and (b) every $k \leq n$ rows have band-union of size $\geq \frac{1}{2} \min(kw, c\varepsilon n)$. Since $a_i \sim U[0, m-w]$ i.i.d. with $m = (1+\varepsilon)n$, (a) is a balls-into-bins bound and (b) follows because k width- w windows placed uniformly in $[0, m]$ cover $\Omega(\min(kw, m))$ points in expectation with Chernoff concentration; both fail with probability $\leq e^{-c_4 w}$. Assume the pattern is good (charged to the final budget).

B.3 Net argument on incompressible and compressible vectors. Fix $\delta, \theta \in (0, 1)$ constants (chosen below) and split S^{n-1} into $\text{Comp}(\delta n, \theta)$ (vectors within Euclidean distance θ of some δn -sparse unit vector) and its complement Incomp .

Incompressible. By the Rudelson–Vershynin structure lemma, every $u \in \text{Incomp}$ has $\geq c_5 n$ coordinates with $|u_i| \geq c_6/\sqrt{n}$; these spread over, by goodness (b), $\kappa(u) \geq \frac{1}{2} \min(c_5 n \cdot w, c\varepsilon n) = \Omega(\varepsilon n)$ active columns. Lemma 1 gives, for each fixed u , $\Pr[\|M^\top u\| \leq \rho\sqrt{w}] \leq (C_0\rho)^{\Omega(\varepsilon n)}$. Take a $\frac{\rho}{2\sigma_{\max}}$ -net $\mathcal{N}$ of Incomp with $|\mathcal{N}| \leq (C/\rho)^n$ (here $\sigma_{\max}(M) \leq C_7(\sqrt{w} + \sqrt{m})$ w.h.p. by the Bai–Yin/ $\sigma_{\max}$ bound, charged to the budget; $\|M^\top \cdot\|$ is $\sigma_{\max}$ -Lipschitz). Choosing $\rho = \rho_0$ a small constant makes $(C_0\rho_0)^{\Omega(\varepsilon n)}|\mathcal{N}| \leq e^{-c_8 \varepsilon n}$, and the Lipschitz net-to-sphere transfer yields $\min_{u \in \text{Incomp}} \|M^\top u\| \geq \frac{1}{2}\rho_0\sqrt{w}$ except with probability $e^{-c_8 \varepsilon n}$.

Compressible. It suffices (by the distance characterization of $\sigma_{\min}$) to bound $\min_{u \in \text{Sparse}(\delta n)} \|M^\top u\|$. A δn -sparse direction activates, by goodness (b), $\kappa(u) \geq \frac{1}{2} \min(\delta n w, c\varepsilon n) = \Omega(w)$ columns (for δ small enough that $\delta n w \leq c\varepsilon n$, i.e. $\delta \leq c\varepsilon/w$, giving $\kappa = \Omega(\delta n w)$; we instead take $k = \Theta(\varepsilon n/w)$ so $\kappa = \Omega(\varepsilon n)$ when that is $\leq w$ and $\Omega(w)$ otherwise). A ρ -net of $\text{Sparse}(k) \cap S^{n-1}$ has $\binom{n}{k}(C/\rho)^k \leq e^{O(k \log(n/k))}$ points. With $\kappa = \Omega(w)$ active columns per point, Lemma 1 gives per-point probability $(C_0\rho)^{\Omega(w)}$; the union is $\leq e^{O(k \log(n/k)) - \Omega(w)} \leq e^{-\Omega(w)}$ once $w \geq c_1 \log n$ and $k \log(n/k) \leq \frac{1}{2} \cdot \Omega(w)$, which holds for $k = O(w/\log n)$. Hence $\min_{u \in \text{Comp}} \|M^\top u\| \geq c_9\sqrt{w}$ except with probability $e^{-c_{10} w}$.

B.4 Assembly. On the good-pattern event (probability $1 - e^{-c_4 w}$) and the two net events (probabilities $1 - e^{-c_8 \varepsilon n}$, $1 - e^{-c_{10} w}$), every $u \in S^{n-1}$ has $\|M^\top u\| \geq c_{11}\sqrt{w}$, i.e. $\sigma_{\min}(M) \geq c_{11}\sqrt{w} \geq c_2$. A union bound over the three failure events gives $\Pr[\sigma_{\min}(M) < c_2] \leq e^{-c_3 \min(\varepsilon n, w)}$, proving Theorem 1. The threshold can be taken $t(\varepsilon, w) = c_{11}\sqrt{w}$ up to the ε -dependence absorbed in the incompressible scale; pinning the sharp $t(\varepsilon, w)$ (in particular the $\varepsilon \rightarrow 0$ collapse, where the incompressible exponent εn and the spectral gap both vanish) is the content of Open Problem 1. $\square$